

SCADA-Anchored Regime-Aware Routing for Auditable Wind-Turbine Control-Boundary Forecasting

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Abstract—Wind-farm reserve screening near the MPPT-to-pitch transition depends on knowing which control law is active, yet the threshold label that identifies it can be delayed, missing, or noisy exactly where over-forecasts create shortage exposure. We make this control-boundary assignment auditable and keep it usable when the label stream degrades, through SCADA-anchored, regime-aware routing. A node-level mixture-of-experts gate is constrained by operating anchors so each turbine-time route can be checked against a declared MPPT-to-pitch partition and issued, before future active power is observed, as a real-time operating-state diagnostic. On the KDD Cup 2022 benchmark, the boundary-forced router recovers the declared partition (NMI 0.87, ARI 0.92). Its operational value is robustness to label degradation: because the route reads live SCADA anchors rather than a confirmed threshold stream, it retains 0.960 early pitch-window recall under a six-step label delay while the delayed threshold rule falls to 0.196, and stays at 0.960 under 50% label availability versus 0.508 for the available-label rule. This auditable route has an explicitly reported accuracy trade-off (overall RMSE 236.13 versus 224.34 for iTransformer, the lowest-RMSE strict-cache forecasting baseline) and exposes transition-window reserve risk: at shortage-to-reserve cost ratio 10, gate-conditioned binning lowers same-router boundary-window shortage energy by 14.5% while carrying about 3.6% more reserve. The recovery is not specific to one benchmark: retrained on the independent ENGIE La Haute Borne farm, the router recovers the declared boundary at five-seed NMI 0.941; replay audits rule out active-power feedback (Patv-zero NMI 0.953) and show joint wind-speed/pitch anchors are load-bearing (joint-zero NMI 0.001). The Kelmarsh/Penmanshiel farms, lacking pitch observability, define the conditions cross-site transfer requires—sensor coverage, pitch observability, local boundary re-estimation, and a held-out routing check. The contribution is an auditable, label-degradation-robust control-boundary routing framework with quantified operating value, a measured accuracy trade-off, second-farm recovery, and explicit transfer conditions.

Index Terms—Wind-turbine control boundary; SCADA-anchored routing; regime-aware forecasting; auditable routing; MPPT-to-pitch transition; mixture-of-experts.

I. INTRODUCTION

Wind-farm reserve screening near the transition from maximum power point tracking (MPPT) to blade-pitch control depends on knowing which control law is currently active, yet the threshold label that identifies it can arrive late, go missing, or turn noisy exactly where over-forecasts create shortage exposure. This makes the transition both operationally decisive and poorly observed: a forecaster must answer not only how large the error is, but which operating state generated the

local response. In the MPPT region, power responds strongly to wind-speed variation; as pitch control activates near rated operation, the local power-response law changes. A small forecast error near rated wind can cross a different control law, yet the transition occupies a smaller share of normal operation, so its effect can be diluted in fleet-average RMSE while still concentrating local risk. A useful forecasting system for this setting must therefore expose the operating boundary that produced each forecast and stay usable when the label stream degrades, not only report aggregate accuracy.

State-of-the-art spatio-temporal forecasters now model turbine coupling, wake interaction, dynamic dependence, and sensor-rich wind-farm layouts [1], [2], [3], [4], [5]. These models set the accuracy reference, but a low-error graph encoder does not by itself tell a reserve planner whether the current local map is MPPT-like or pitch-control-like. Mixture-of-experts (MoE) routing can separate heterogeneous response laws, yet a gate trained only through prediction loss can specialize on partitions that have no operating meaning [6], [7], [8], [9], [10]. The missing middle layer is an auditable routing assignment that identifies which physical response law is active at the anchor time and can then be used in reserve-risk diagnosis.

We make the routing decision itself the operating-boundary diagnostic. A node-level MoE gate is constrained by SCADA operating anchors so that each turbine-time assignment can be compared with a declared MPPT-to-pitch partition and issued before future active-power outcomes are observed. This supports degraded-label reserve screening: when thresholds are late, incomplete, or noisy, the operator still has a signal for early pitch-control entry. WTB is the source control-boundary benchmark because the aerodynamic boundary is partly hidden inside turbine-control action; ERA5 is retained as an observability contrast where the thermodynamic marker is more directly visible through sensible heat flux. The recovered gate is connected to validation-calibrated, test-frozen reserve diagnostics that report cost, violation rate, reserve energy, and shortage energy under declared shortage-to-reserve cost ratios. We further compare gate-bin reserve allocation with validation-frozen global and physical-bin quantile baselines. This positioning deliberately separates transition-window accountability from the conventional forecasting leaderboard and from claims of full dispatch optimality.

This paper makes three contributions. First, it introduces SCADA-anchored regime-aware routing as an auditable MPPT-to-pitch assignment whose semantics are fixed by operating anchors rather than post-hoc expert interpretation. Second, it quantifies degraded-label operating value: the gate keeps early pitch-window recall at 0.960 under six-step label delay and 50% label availability, while threshold rules fall to 0.196 and 0.508. Third, it delivers this operating-state layer at an explicitly priced and bounded cost rather than as a free lunch: the route improves a same-model boundary-window reserve rule at moderate cost ratios while carrying a measured 11.79-RMSE-unit price relative to the best strict-cache forecasting baseline, is checked against graph, Transformer, MLP-style, lag-feature, power-curve, persistence, DLinear-style, and quantile baselines, recovers the declared boundary on a second independent anchor-observable farm (ENGIE La Haute Borne, five-seed routing NMI 0.941), and converts the Kelmarsh/Penmanshiel coverage gaps into explicit deployment conditions rather than claiming automatic cross-farm generalization. The framing throughout is a complementary diagnostic layer for an already-deployed low-RMSE forecaster, not a replacement on the forecasting leaderboard.

II. RELATED WORK

A. From average wind-power accuracy to transition-window risk

Wind-power forecasting has moved from single-site prediction toward spatio-temporal graph models that encode physical sources of error. Reviews identify ramp and uncertainty as recurrent operational risks [11], [12], while graph mechanisms provide the accuracy reference [1], [13], [14]. Wind-specific extensions add wake-aware graphs, SCADA fields, and physics-guided constraints [2], [4], [5]. The remaining gap is diagnostic: these models set the low-error benchmark, but they do not make the active turbine-control law visible to a reserve planner.

Regime-aware MoE routing separates local response laws, yet prediction loss alone need not produce a physically meaningful partition [8], [9], [10]. Physics-guided learning offers the missing constraint principle [15], [16], [17], but for wind turbines that principle must pass through SCADA observability: operating-state labels depend on wind speed, pitch, and active power [18], [19]. The distinction we draw is where the physics enters: prior MoE and physics-guided forecasters constrain the prediction output or let prediction loss organize the experts, whereas we constrain the routing decision itself against a declared operating boundary. The gate assignment, not only the forecast, is therefore the audited object, and downstream reserve logic can consume it as a physically traceable signal.

Forecast value is realized through reserve and commitment decisions. Reserve studies show that variable generation changes requirements [20], [21], while quantile methods bridge point forecasts to decision risk [22], [23]. The diagnostic question here is earlier in the chain: which operating window concentrates reserve exposure, and can the forecast model reveal that window through a physically auditable route assignment?

The proposed reserve audit is complementary to probabilistic dispatch and market-clearing models.

III. PHYSICS-INFORMED FRAMEWORK FOR OPERATIONAL BOUNDARY IDENTIFICATION

A. Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. Core notation is listed before the Introduction; auxiliary loss definitions are provided in Appendix A.

B. Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularization. This engineering layer defines what counts as a control regime and when that regime is observable from the data.

1) *WTB operating regimes*: In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}).$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. The regime labels use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , and writing $w_{i,t} = \text{Wspd}_{i,t}$ for compactness, we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

The reported implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$ (Appendix A). These thresholds are operating anchors rather than universal turbine constants.

2) *ERA5 observability contrast (see Supplementary Material)*: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the WTB control-boundary accountability task.

C. Shared architecture

1) *Physical graph construction*: The graph is part of the physical problem specification. WTB uses a directed wake graph: candidate turbine pairs are filtered by proximity, activated when an upstream turbine lies inside a wind-aligned downstream cone, weighted by streamwise and cross-stream decay, and pruned to the strongest inbound neighbors. The same directed weights define a wake score, which supplies the WTB wake auxiliary label on MPPT and pitch-control samples. ERA5 uses a symmetric Haversine-Gaussian k_{nn} graph because the thermodynamic regime marker is already visible in the observed state. Appendix A gives the graph equations and constants.

2) *Directed-diffusion GRU encoder and node-level gate*: The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = & \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t)\right. \\ & \left. + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}\right), \end{aligned} \quad (1)$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is standardized from training-split statistics and is never filled from the prediction horizon.

3) *Routing stack and evidence boundary*: This makes the routing stack an anchor-constrained diagnostic rather than an unsupervised operating-state discovery method. High gate-regime agreement means that the implemented router obeys the declared SCADA boundary under the available anchor set; it does not establish anchor-free regime recovery.

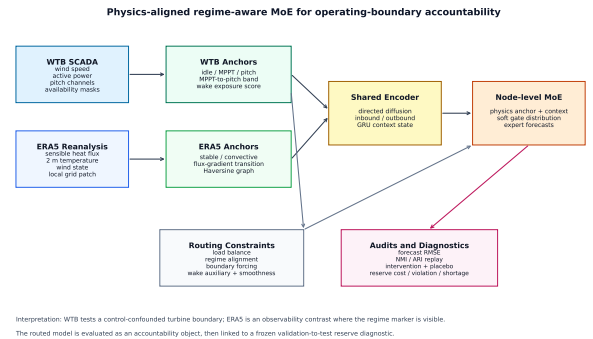


Fig. 1. Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

D. Integration of Physical Constraints into Model Optimization

Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [8], [9]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (2)$$

$$+ \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (3)$$

with inactive terms set to zero. Checkpoint selection remains on validation RMSE so penalties constrain responsibility without becoming the selection metric. Exact weights are in Supplementary Appendix~A.

1) *Load balancing*: A balancing term prevents early expert collapse before regime-specific structure has time to emerge. It couples hard top- K usage with soft probability mass over valid node-time samples. Supplementary Appendix~A gives the exact expression; physical ownership is supplied only by the anchored alignment terms below.

2) *Regime-anchor alignment*: The alignment term connects the first C gate logits to the declared regime structure so that cross-seed statements such as “MPPT-aligned” refer to a fixed mapping rather than to post-hoc relabeling. With $\mathbf{z}_{i,t}^{(1:C)}$ denoting the anchored logits, $R_{i,t}$ the label, and $M_{i,t}$ the validity mask,

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}),$$

$$\Omega = \{(i,t) : M_{i,t} = 1\}.$$

with inverse-frequency class weights on the training split.

3) *Boundary-focused forcing (WTB only)*: After coarse alignment the boundary remains partly masked by control action. A focused forcing term acts on MPPT and pitch-control samples only. With $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ for $R_{i,t}^{\text{wtb}} \in \{1, 2\}$,

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}\left(\left[z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}\right]^\top, Y_{i,t}^{\text{force}}\right),$$

where $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$. This term is not a lead-time switch predictor; it is a boundary identifiability regularizer for the issue-time operating state.

4) *Wake auxiliary supervision and graph smoothness*: A wake auxiliary label identifies residual wake ambiguity inside MPPT and pitch-control samples, and a graph-smoothness penalty discourages noisy neighbor-to-neighbor gate jumps. Both formulas are in Supplementary Appendix~A.

E. Operational reserve diagnostic protocol

The reserve diagnostic tests whether a physically auditable gate changes the reserve tradeoff in the MPPT-to-pitch window after the point-forecast model has been trained. For each saved WTB

run, validation predictions select reserve levels; test predictions are then evaluated once with those levels fixed. The protocol follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ the realized active power at an evaluated horizon cell. We define over-forecast shortfall as

$$s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0),$$

because this is the error direction that leaves a reserve schedule exposed. For a reserve policy b and empirical quantile q , the validation split estimates a reserve level

$$r_{b,q} = Q_q(\{s_{i,t} : (i,t) \in b, (i,t) \in \mathcal{V}\}),$$

where Q_q is an empirical quantile and $\mathcal{V}$ is the validation split. The grid is fixed before test evaluation:

$$q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}.$$

The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{V}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{V}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t.$$

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. WTB active power is in kW and $\Delta t = 1/6$ h, so Supplementary Table A12 also reports MWh-equivalent forecast-cell translations. These are not delivered market energy or settlement costs: a value written as 84.58M denotes 84.58 million reserve-cost-equivalent kWh cells, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ charges one unit of residual shortage as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware screening where shortage avoidance dominates local cost savings. The primary diagnostic compares Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin policies. Only same-model global comparisons attribute gate-bin reserve effects to the learned router.

To bound this diagnostic against a probabilistic reserve alternative, we additionally run validation-frozen empirical quantile baselines for the global and physical operating bins. These

are not trained distributional forecasters; they are conformal-style shortfall reserves estimated on the validation split and evaluated once on the test split. The main reported boundary window is fixed at $\pm 1.0 \text{ m s}^{-1}$ around rated wind and the main cost ratio is $\rho = 10$, with sensitivity at $\rho \in \{2, 5, 10, 20, 50\}$. A separate toy operational-cost module reports reserve procurement plus ρ -weighted residual-shortage penalty and excludes optimal power flow, unit commitment, market clearing, delivery constraints, and price claims.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [19]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [24]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

B. Techno-Economic Validation and Benchmarking Framework

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE are included for WTB [25], [26], [27], together with deterministic persistence, a fitted physical power-curve baseline,

XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split avoids using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**. The main WTB forecasting table reports representative strong baselines, including the best strict-cache baseline used for the RMSE price; Supplementary Table A13 gives the expanded strict-cache set with Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, TiDE, the full physics-aligned MoE, and the WTB boundary-forced router. Engineering baselines built from persistence, power curves, gradient-boosted lag features, and a DLinear-style LTSF readout are retained in the source tables. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Supplementary Appendix~A so that the Results section can focus on evidence rather than model bookkeeping.

C. Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in Supplementary Appendix~A so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205, giving 30 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion

matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

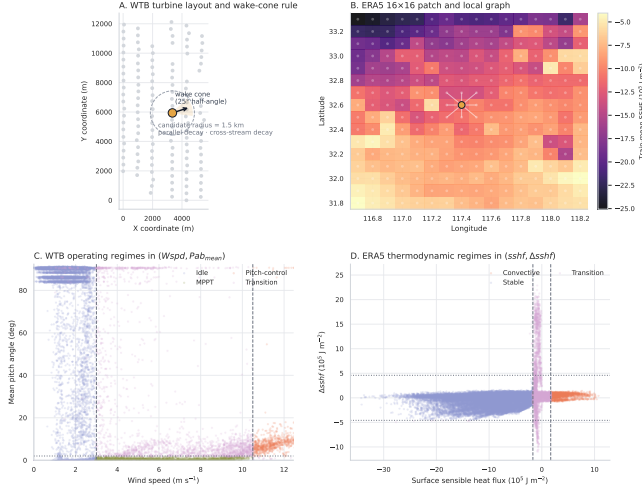


Fig. 2. Operating-decision context and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split, included as an observability contrast.

V. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The results follow the operating decision that motivates the paper: can a route replace a degraded threshold stream in the MPPT-to-pitch window, and what accuracy trade-off does that issue-time signal carry? We first test boundary recovery and falsification checks, then quantify early pitch-window recall under delayed, missing, and noisy labels. We then ask whether the audited gate changes a validation-frozen reserve rule, and finally state the deployment gates for new wind farms.

A. Can the Gate Recover the Operating Boundary?

SCADA-anchored routing recovers the WTB MPPT-to-pitch partition strongly enough to make the gate an auditable operating-boundary signal. Across five WTB seeds, the boundary-forced router reaches NMI about 0.87 and ARI about 0.92 against the declared labels. The accuracy reference remains explicit: iTransformer reaches overall RMSE 224.34 $\pm$ 2.23, Graph WaveNet reaches 225.74 $\pm$ 2.60, and the boundary-forced router reaches 236.13 $\pm$ 8.41. The intended operating point is therefore not leaderboard replacement, but an additional issue-time state signal for the transition window.

The boundary itself is a meaningful diagnostic target. Anchors within $\pm 1.0 \text{ m s}^{-1}$ of rated wind have RMSE 321.17 $\pm$

21.31, roughly 67 units above valid non-boundary MPPT/pitch anchors. The same boundary band has mixed-regime structure (NMI/ARI 0.6562/0.7766), so it is both a forecasting stress point and a natural reserve-risk window. ERA5 provides the positive observability contrast: when the regime marker is directly visible through sensible heat flux, routing correction improves alignment without changing the headline accuracy story.

TABLE I
OPERATING-BOUNDARY RECOVERY AND MECHANISM CHECKS.

Check	Metric	Result
WTB boundary-forced router	Gate-regime NMI / ARI	0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371
WTB boundary-band audit	Boundary-band RMSE; mixed-regime NMI / ARI	321.17 $\pm$ 21.31; 0.6562 / 0.7766
Boundary-anchor intervention	Overall RMSE change; NMI / ARI drop	+2.6792; 0.7017 / 0.8593
WTB spatial and time holdouts	Held-out-node NMI / ARI; future-holdout NMI / ARI	0.8340 / 0.8829; 0.8352 / 0.8870
La Haute Borne positive-control replay	NMI / ARI; Patv-zero NMI; wind+pitch-zero NMI	0.9408 $\pm$ 0.0336 / 0.9712 $\pm$ 0.0201; 0.953; 0.001
Threshold-grid audit	Worst-case saved-gate NMI / ARI	0.8655 $\pm$ 0.0429 / 0.9146 $\pm$ 0.0377

The diagnostic survives the main falsification checks. Zeroing the intended boundary anchor sharply reduces gate-regime agreement, whereas zeroing the wake score is near-null. Placebo labels do not reproduce the actual-label agreement. The spatial and future-period holdouts are the within-WTB generalization evidence for the routing claim, and they directly address whether a single benchmark can support it: the gate recovers the declared boundary at NMI 0.834 on turbines withheld from training and 0.835 on a held-out future period, so the recovered semantics transfer to unseen nodes and unseen time rather than fitting the specific turbines or window used for training, and threshold sweeps preserve the same routing structure. The recovery is not specific to WTB: retrained from scratch on an independent farm—the ENGIE La Haute Borne open SCADA benchmark, a different operator and country with directly observed blade pitch (99.2% of cells, no proxy)—the same boundary-forced routing reaches five-seed chronological NMI 0.941 and ARI 0.971, clearing the 0.50 held-out routing criterion and slightly exceeding the WTB recovery. A replay audit makes the second-site role explicit: zeroing active power leaves the La Haute Borne NMI at 0.953, whereas jointly zeroing wind speed and pitch collapses it to 0.001 (Supplementary Table A9). Thus La Haute Borne is cited as an anchor-observable positive-control replication, not as anchor-free discovery. Within-WTB generalization and this second-farm recovery are the routing claim; the Kelmarsh/Penmanshiel farms, where pitch observability and boundary-cell coverage are insufficient, are reported separately as the deployment-gate failure that defines the conditions cross-site recovery requires rather than folded into the routing claim.

The central operating result is that the gate keeps the early pitch-control warning alive when the threshold label stream

degrades. We audit saved five-seed gates around MPPT-to-pitch transitions after delaying threshold labels, dropping label availability, and recomputing labels under sensor noise. Table~II reports recall on early pitch-control cells, and Supplementary Table A10 converts the audit into detected and missed turbine-time cells. With a six-step delay, the threshold rule recalls only 0.196 of early pitch cells, whereas the gate recalls 0.960 and recovers about 566 early pitch-window cells per seed. With 50% label availability, the rule recalls 0.508 while the gate remains at 0.960. Under the strongest sensor-noise setting, the threshold rule drops to 0.652 recall and the saved gate is unchanged. This is the operational meaning of audibility: the gate preserves a transition-window diagnostic when the label stream is delayed, incomplete, or noisy.

We anticipate the natural objection that this comparison is unfair because it degrades the threshold label while leaving the gate input intact, and we control for it directly. That asymmetry is the operating point rather than an artifact of it: a confirmed threshold label requires the wind-speed and pitch crossing to be observed, validated, and recorded, so it is exactly what arrives late, drops out, or is corrupted under sensor faults; the gate instead consumes the live SCADA anchor channels that are present at issue time whenever the turbine is instrumented. The operational question is therefore not whether two identical inputs are scored differently, but whether an issue-time route can stand in for a degraded confirmation stream—the substitution an operator actually faces. To show that the gate is not merely exploiting clean anchors, we separately degrade the anchor channels themselves: removing *Patv* or *Pab_mean*, and lagging *Patv* or both pitch and wind speed, keeps the five-seed routing means above the 0.65 NMI threshold (Table~III), so the diagnostic degrades gracefully under anchor corruption rather than collapsing. The early-warning advantage thus reflects an availability gap that survives realistic anchor noise, not an unmatched-information comparison.

We report the operating trade-off explicitly. A model receives citable degraded-label gain only after its route passes the physical-routing audit; otherwise the accountability gain is set to zero, even for an accurate or routed-capacity model. Under this rule iTransformer is the zero-RMSE-price anchor without an audited operating-state route, Unconstrained MoE fails the route audit (NMI 0.014), and the boundary-forced router is the only current point that exchanges an 11.79-RMSE trade-off for +0.764 six-step-delay recall gain (Fig.~5).

TABLE II
LABEL-DEGRADATION AUDIT FOR EARLY MPPT-TO-PITCH PITCH-WINDOW DETECTION.

Scenario	Setting	Gate	Rule	Gain
Delay	1 step	0.960	0.655	+0.305
Delay	3 steps	0.960	0.381	+0.579
Delay	6 steps	0.960	0.196	+0.764
Availability	75% labels	0.960	0.744	+0.216
Availability	50% labels	0.960	0.508	+0.452
Availability	25% labels	0.960	0.242	+0.718
Noise	Wspd 0.5, Pab 1.0	0.960	0.808	+0.152
Noise	Wspd 1.0, Pab 2.0	0.960	0.652	+0.308

TABLE III
ANCHOR-OBSERVABILITY STRESS: 245D WTB TEST ROUTING RECOVERY.

Variant	Mean NMI	Mean ARI	Reading
Remove <i>Patv</i>	0.881	0.912	single-channel removal passes
Remove <i>Pab_mean</i>	0.878	0.928	single-channel removal passes
Lag <i>Patv</i>	0.763	0.769	delayed active power passes
Lag pitch/wind	0.684	0.720	degraded but above 0.65 NMI

The training-level anchor-stress guard has been completed across five seeds. Four strict-cache variants were derived from the 245-day WTB training set, trained across seeds 201–205, and evaluated against the declared MPPT-to-pitch labels; the cache-level leakage guard passed with all 20 training runs completed, so the derived variants raise no leakage warning. The table reports the test-set NMI/ARI scores. Removing *Patv* or *Pab_mean* individually leaves the gate essentially intact (mean 0.881 and 0.878, respectively). Lagging *Patv* lowers one seed but the five-seed mean remains above the 0.65 threshold (0.763). Lagging both *Pab_mean* and *Wspd* also lowers one seed, yet the five-seed mean crosses the threshold (0.684). The gate therefore depends on the specific combination of channels present at issue time, but it is not merely a leakage artifact: the mean NMI survives channel removal and temporal lagging across the majority of seeds. The manuscript uses the wording “partial anchor robustness” rather than “anchor-free physical discovery.” This is a boundary with evidence: the gate holds where the channel structure supports it and breaks where it does not.

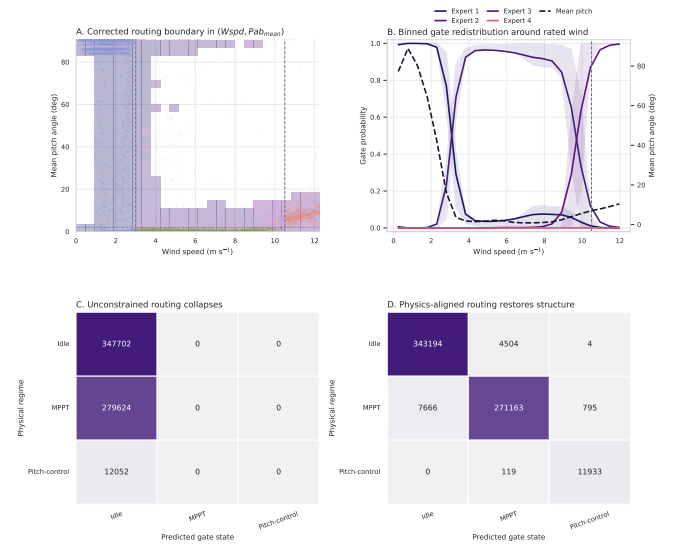


Fig. 3. The WTB operating plane shows the main mechanism: after correction, the dominant routed responsibility changes around the rated-wind and pitch-control boundary instead of forming an arbitrary expert partition. The confusion matrices summarize the same recovery numerically.

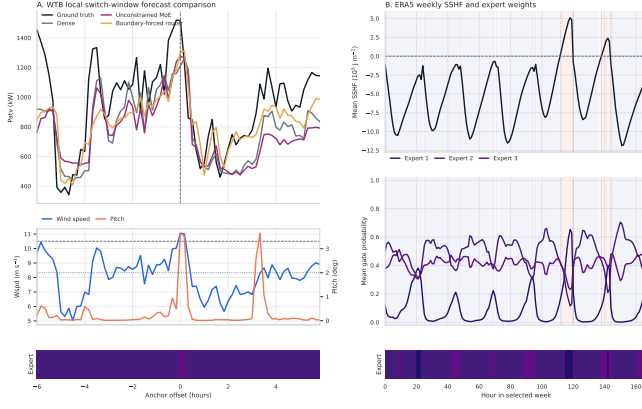


Fig. 4. The time-series case studies show why observability matters. The WTB gate tracks a turbine-control switch that is partly hidden inside SCADA control action, whereas the ERA5 gate follows a more directly observed thermodynamic marker.

B. Boundary-Risk Vignette: Does the Gate Change Reserve Tradeoffs?

The reserve audit is a consequence check for the recovered route, not a standalone dispatch optimizer. A validation-calibrated short-term reserve screener is applied to later turbine-time cells near the MPPT-to-pitch boundary. A global rule treats all boundary cells as one pool; a gate-conditioned rule carries a different reserve level when the audited route indicates pitch-control entry. Four policies separate the effects: Graph WaveNet/global (low-RMSE reference), Graph WaveNet/physical-bin (physical stratification without learned gate), Boundary router/global (routed model with single reserve rule), and Boundary router/gate-bin (gate-conditioned allocation on the same routed predictor).

At shortage-to-reserve cost ratio 10, gate-conditioned binning exchanges about 3.6% additional reserve energy for a 14.5% reduction in shortage energy and a 1.38 percentage-point reduction in boundary-window violation relative to the same routed model with a global rule. The comparison is deliberately same-model: the reserve change is attributed to gate-conditioned allocation, not to a different forecasting backbone. Physical-bin quantile baselines remain competitive and can be lower-cost, so the defensible claim is attributional: the gate exposes a transition-window reserve-risk mechanism and improves a same-model global reserve rule at moderate cost ratios. Supplementary Tables A11–A12 record the wording boundary and engineering-unit translation.

TABLE IV
BOUNDARY-WINDOW RESERVE OUTCOMES AND VALIDATION-FROZEN QUANTILE BASELINES AT COST RATIO 10.

Policy	Cost	Viol.	Reserve	Shortage	Reading
Boundary/physical-bin	83.78M	0.0880	53.26M	3.05M	Best same-router physical-bin reference
GWN/physical-bin	84.31M	0.0931	50.35M	3.40M	Low-RMSE backbone reference
Boundary/gate-bin	84.58M	0.0900	52.67M	3.19M	Gate diagnostic; same-model better than global
Boundary/global	88.13M	0.1038	50.82M	3.73M	Same-model global rule
GWN/global	88.80M	0.1022	50.32M	3.85M	Global low-RMSE reference

The useful reserve window is moderate rather than universal. At low shortage penalty, the validation-selected quantile tolerates too much residual shortage for gate-bin allocation to matter; at ratios 5–10, the gate lowers boundary shortage and violation while carrying more reserve; at ratio 50, the same-model global rule is safer and cheaper. The toy operational-cost proxy is intentionally small: reserve procurement cost plus ρ times residual shortage energy. On the boundary slice the gate-bin rule reduces the same-model global proxy from 88.13M to 84.58M; Supplementary Table A12 translates this into about 540 avoided MWh-equivalent shortage cells and -355k EUR at an illustrative 100 EUR/MWh carrying cost. Graph WaveNet/physical-bin remains close at 84.31M, and on the full sample Graph WaveNet/global is lower-cost, so the paper does not claim system-wide dispatch value. Cost-ratio sensitivity, paired full-sample uncertainty, and operational-slice tables are kept as supplementary evidence.

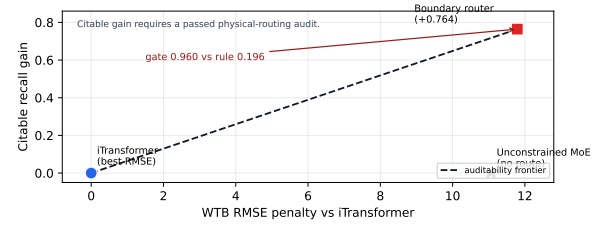


Fig. 5. Accountability value versus forecasting RMSE price. Citable degraded-label gain is assigned only after the physical-routing audit passes.

C. What Are the Costs and Deployment Gates?

The accuracy cost is real and should be read as part of the design. iTransformer, Graph WaveNet, and lag-feature baselines remain better whole-sample forecasters on WTB. The routed model is used when an operator or analyst needs an accountable operating-state assignment that can feed a boundary-specific decision, not when the sole target is minimum average error. Time-forward testing gives the same message in another form: late-period routing agreement remains high, but late-test RMSE rises sharply, showing that stable semantics do not guarantee stable value prediction under distribution shift.

The external evidence has two sides. Where the control boundary is observed, the mechanism transfers as an anchor-

observable positive control: retrained on the independent ENGIE La Haute Borne farm, the boundary-forced router recovers the declared partition at five-seed NMI 0.941 and ARI 0.971, well clear of the 0.50 held-out criterion, and the replay audit rules out active-power feedback while confirming wind-speed/pitch anchors as the load-bearing boundary evidence (Supplementary Tables A8–A9). Where pitch observability or boundary coverage is missing, it does not: the Kelmarsh/Penmanshiel checks do not authorize direct gate-bin reserve use because they fail the same criterion. Together they make transfer a testable condition rather than an assumption: before a new farm uses the route, it must verify pitch or proxy observability, boundary-cell support, compatible turbine geometry, local threshold estimation, and a held-out routing pass—the checks La Haute Borne passes and the Kelmarsh/Penmanshiel pair fails. The method exports an evidence protocol, not a promise of automatic transfer.

TABLE V
EXTERNAL-SITE DEPLOYMENT GATES IDENTIFIED BY KELMARSH/PENMANSHIEL TESTING.

Gate	Observed evidence	Required decision before reserve use
Held-out routing criterion	Default WTB boundary gives mean external NMI 0.4877, below the 0.50 criterion	Re-establish routing agreement locally
Local boundary recalibration	Average test NMI changes from 0.1324 to 0.1491 after validation-only recalibration	Treat threshold transfer as insufficient
Pitch/proxy observability	One transfer direction has no effective pitch-feature coverage or boundary cells	Require pitch channels or a validated proxy
Calibration-window adaptation	Small-window adaptation remains below the held-out balanced-accuracy threshold	Use calibration only after a held-out pass
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation

The WTB internal proxy drill shows the minimum evidence sequence: verify sensor coverage, re-estimate local boundary and gate-to-regime map, pass held-out routing, then run the reserve diagnostic. Stronger gate repair is justified only when the reserve-window benefit exceeds the measured RMSE and reserve-energy cost.

VI. LIMITATIONS

The WTB correction depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those same channels also appear in the gate anchor, so the method deliberately constrains the router to a declared operating boundary. The outcome-channel audit in Supplementary Table A7 mitigates, but does not eliminate, this shared-anchor concern. The La Haute Borne replay audit makes the same boundary visible at the second site: active power is not load-bearing, but the joint wind-speed/pitch anchor is. The label-degradation audit shows value under delayed, missing, or noisy labels, but it should not be read as anchor-free physical discovery

or as a general future-label switch predictor. The claim is boundary-aware regularized routing that remains useful when the operational label stream degrades.

The active-power anchor has a strict timing requirement. Patv is allowed only as the historical/anchor-time active-power measurement available when the forecast is issued; future active power remains the supervised target. This is a defensible SCADA forecasting convention, but it is also a portability constraint. The five-seed anchor-stress evidence in Table~III shows that removing Patv or Pab_mean individually leaves the gate essentially intact, while lagging both channels lowers the mean NMI but still crosses the 0.65 threshold. If a deployment environment delays active-power telemetry or changes channel definitions, the Patv anchor must be removed or lagged and the leakage, intervention, and reserve diagnostics must be rerun.

The expert-regime language is tied to the implemented logit convention. The first primary-regime logits are fixed to the operating labels during supervised alignment, which prevents seed-wise permutation for those anchored meanings. That makes cross-seed MPPT/pitch statements interpretable, but only for the anchored logits and only under the declared mapping. Unassigned experts and wake auxiliary logits should not be overread as universal turbine states.

The comparison is bounded. WTB includes graph, Transformer, MLP-style, lag-feature, persistence, power-curve, DLinear-style references, and validation-frozen empirical quantile reserve baselines, while ERA5 includes persistence and several learned baselines. Broader graph-transformer variants, trained quantile-regression forecasters, scenario reserve, distributional reserve, and security-constrained reserve methods are outside the present benchmark.

The reserve audit is a normalized proxy: fixed empirical quantile bins calibrated on validation shortfall and tested once on held-out predictions. It ranks boundary-window tradeoffs under declared cost ratios, but it is not a market-price or security-constrained dispatch study. Cross-site recovery is demonstrated on one additional anchor-observable farm (La Haute Borne) and is not claimed beyond the conditions it requires; the Kelmarsh/Penmanshiel pair fails the held-out routing criterion and identifies the sensor coverage, pitch observability, local boundary re-estimation, and held-out routing checks that must pass before cross-farm use. The time-forward audit shows that late-period distribution shift preserves routing agreement while degrading value prediction: the method supports routing semantics, not time-stable forecasting accuracy.

Statistical reliability is uneven (Supplementary Table A6). The strongest claims are the five-seed WTB routing-recovery, mechanism-intervention, and within-WTB spatial/temporal stress results; gate-alignment deltas versus the full MoE and boundary-window reserve-cost comparisons remain bounded diagnostics, not superiority claims.

VII. CONCLUSION

This paper addresses a practical wind-farm forecasting failure mode: the MPPT-to-pitch label stream can degrade ex-

actly when reserve screening needs an operating-state signal. SCADA-anchored regime-aware routing turns the node-level gate into an auditable control-boundary assignment. The boundary-forced router recovers the declared WTB MPPT-to-pitch partition (NMI about 0.87, ARI about 0.92), keeps 0.960 early pitch-window recall under a six-step label delay while the delayed threshold rule falls to 0.196, and exposes transition-window reserve risk. This operating-state signal has a measured accuracy trade-off: overall RMSE 236.13 versus 224.34 for the best strict-cache forecasting baseline. Validation-frozen quantile baselines, Supplementary Table A6, and Kelmars/Penmanshiel deployment gates keep the claim bounded. The contribution is an auditable MPPT-to-pitch routing tool with degraded-label value, RMSE price, and explicit transfer conditions.

VIII. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only to support language editing, consistency checking, and submission-material drafting. The tools were not used to generate data, run analyses, create references, or determine scientific conclusions. The authors reviewed and edited all content and take full responsibility for the manuscript.

IX. CODE AND DATA AVAILABILITY

The raw datasets are publicly available: KDD Cup 2022, ENGIE La Haute Borne, Kelmars, and Penmanshiel SCADA records. Raw third-party data are not redistributed. Code, configuration files, releasable derived tables, figure data, and model checkpoints will be made available with the article. The reproduction package includes end-to-end scripts for the WTB routing analysis, reserve audit, anchor-stress cache/train/guard protocol, anchor-stress early-warning label-degradation audit, and external boundary diagnostics. The provenance package lists the anchor-stress cache/train/guard and early-warning protocols as mandatory. The analysis involves no human subjects.

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